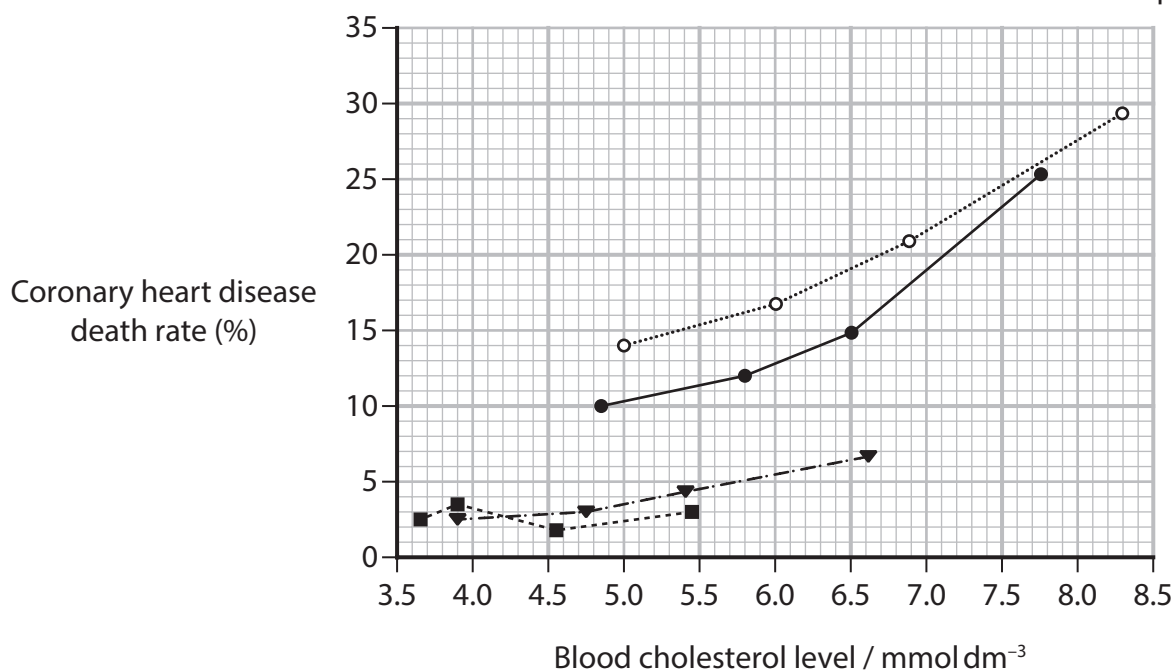


6 A high blood cholesterol level is associated with an increased risk of cardiovascular disease (CVD).

(a) The graph shows the relationship between blood cholesterol levels and the coronary heart disease death rate in men in four parts of the world.

Key:

-○..... Northern Europe
- United States
- ▼--- Mediterranean
- Japan



Describe **two** conclusions that can be made about coronary heart disease death rate in these parts of the world.

(2)

1

.....

.....

2

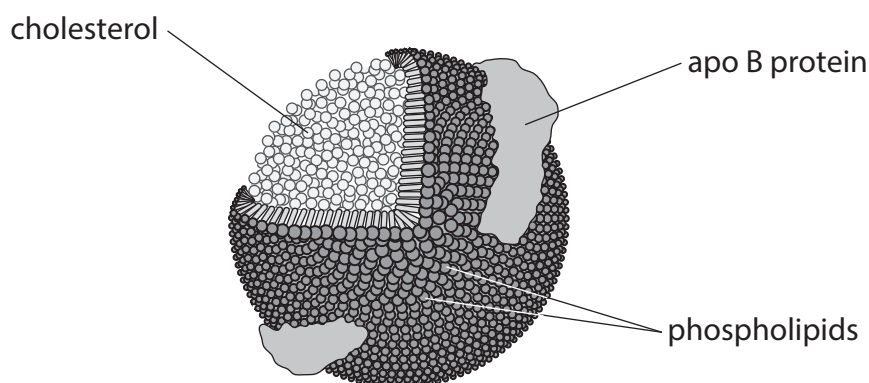
.....

.....



- (b) Cholesterol can be transported in the blood as low-density lipoproteins (LDLs) that are made in the liver.

The diagram shows a LDL.



Statins can be used to reduce blood cholesterol levels.

Statins reduce cholesterol synthesis, reduce the synthesis of apo B protein and increase the number of LDL receptors on the surface of liver cells.

- (i) Statins can cause side effects.

Give **two** side effects of taking statins.

(1)

1

2

- (ii) Liver cells take up cholesterol by endocytosis.

Explain how an increase in these LDL receptors will help to reduce blood cholesterol levels.

(2)

.....

.....

.....

.....

.....

.....

.....

.....

.....



(c) Familial hypercholesterolemia (FH) is a genetic disorder that results in high levels of LDLs in the blood.

(i) In the UK, approximately 270 000 people have FH.

The disorder affects 1 in 250 people in the UK.

Calculate the population size of the UK.

Express your answer in standard form.

(2)

Answer

(ii) The disorder can be caused by a mutation in the gene that codes for LDL receptors.

Suggest why statins are less effective in treating people who are homozygous for this disorder.

(2)

.....

.....

.....

.....

.....

.....



- (iii) Another drug, used to treat FH, is a short antisense strand of DNA from the apo B gene.

Suggest how this antisense strand of DNA would reduce the levels of apo B protein.

(2)

(Total for Question 6 = 11 marks)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

